Machine Anomaly Response Manual

Operational guidance for identifying machine issues and applying standard recovery actions.

1. Purpose

This manual provides standard operating guidance for responding to machine anomalies across the production line. It is intended for operators and first-line supervisors who need to identify the issue, understand the affected parameters, and apply the approved recovery action in a consistent and safe manner. The procedures in this document are first-response instructions only and should be followed before escalating to maintenance or engineering when the anomaly persists.

2. Standard Recovery Actions

The following recovery actions are the only standard actions referenced in this manual. Operators must apply only the action stated for the specific anomaly and must confirm that the machine is in a safe condition before restarting equipment.

- Stop machine, start machine
- Clear anomaly, stop machine, start machine
- Clear anomaly

3. Machine-Specific Operating Instructions

3.1 Scale

The scale is used to verify ingredient weight and dosing stability before product transfer to the next process stage. Reliable readings are essential to maintain recipe accuracy and batch consistency.

3.1.1 Drift

Affected parameters: weight_kg, stability

Issue description: Drift occurs when the measured value gradually moves away from the expected reference value. This can cause underfilling, overfilling, or unstable weighing results across the batch.

Required operator action:

1. Clear the anomaly from the operator interface.
2. Stop the machine.
3. Start the machine and confirm that the reading is stable before resuming operation.

Note: If instability remains after restart, the scale should be removed from normal operation and inspected for calibration or mechanical alignment issues.

3.2 Mixer

The mixer combines ingredients and maintains uniform blending before thermal treatment or downstream processing. Proper blade movement and temperature control are required to achieve the target mixture consistency.

3.2.1 Blade Jam

Affected parameters: rpm, torque_nm

Issue description: A blade jam indicates that the mixing assembly is obstructed or unable to rotate freely. This usually appears as reduced speed and increased torque demand.

Required operator action:

1. Stop the machine.
2. Start the machine and verify that normal blade movement has resumed.

Caution: Do not continue operation if abnormal torque or vibration is still present after restart.

3.2.2 Overheating

Affected parameters: temperature

Issue description: Overheating indicates that the mixer temperature has exceeded the normal operating range. Typical causes include excessive mechanical load, friction, or insufficient cooling.

Required operator action:

1. Clear the anomaly from the operator interface.
2. Stop the machine immediately.
3. Start the machine back to operation.

Warning: Do not restart the mixer under this condition unless specifically instructed by maintenance or supervisory personnel.

3.3 Mix Cooker

The mix cooker heats and agitates the product to achieve the required thermal and mixing profile. Stable temperature and stirring performance are critical to prevent product damage.

3.3.1 Burn

Affected parameters: temperature

Issue description: Burn indicates overheating severe enough to scorch the product or damage the heated surface.

Required operator action:

1. Clear the anomaly.
2. Stop the machine.

3. Start the machine back to operation.

Warning: Continued operation can compromise product quality and may damage the vessel surface.

3.3.2 Stall

Affected parameters: stirring_rpm

Issue description: Stall occurs when the stirring mechanism stops or slows unexpectedly, reducing or preventing proper agitation.

Required operator action:

1. Stop the machine.
2. Confirm the agitator is fully stationary.
3. Start the machine and verify that stirring speed returns to normal.

Note: If repeated stalling occurs, the unit should be inspected before the next batch is processed.

3.4 Pasteurizer

The pasteurizer controls thermal treatment to ensure the product reaches and maintains the required temperature profile. Both temperature and pressure must remain within control limits.

3.4.1 Overheating

Affected parameters: temperature

Issue description: Overheating means the process temperature has exceeded the safe control range.

Required operator action:

1. Clear the anomaly.
2. Stop the machine.
3. Start the machine back to operation.

Warning: Product integrity may be affected if the cycle continues above the allowed temperature range.

3.4.2 Pressure Drop

Affected parameters: pressure

Issue description: Pressure drop indicates that system pressure is below the expected operating value, which may reflect leakage, restriction, or pump inefficiency.

Required operator action:

1. Clear the anomaly.
2. Stop the machine.
3. Start the machine.

Note: If the pressure condition returns, escalate for maintenance inspection.

3.5 Homogenizer

The homogenizer reduces particle size and stabilizes product texture by maintaining controlled pressure through the homogenizing stage.

3.5.1 Valve Failure

Affected parameters: pressure_bar, particle_size_um

Issue description: Valve failure disrupts pressure regulation and can lead to non-uniform product texture.

Required operator action:

1. Clear the anomaly.
2. Stop the machine.
3. Start the machine.

Caution: Do not continue production if abnormal pressure persists.

3.6 Aging Tank

The aging tank holds the mix under controlled conditions to stabilize product structure before freezing. Temperature and fill level must remain within specification.

3.6.1 Temperature Rise

Affected parameters: temperature

Issue description: Temperature rise indicates loss of cooling control during the aging phase.

Required operator action:

1. Clear the anomaly.
2. Stop the machine.
3. Start the machine back to operation.

Warning: Do not continue the batch if product temperature cannot be maintained.

3.6.2 Overflow

Affected parameters: level_pct

Issue description: Overflow occurs when the tank level exceeds the normal limit, creating spill and contamination risk.

Required operator action:

1. Clear the anomaly.
2. Stop the machine.
3. Start the machine back to operation.

Caution: Verify surrounding surfaces and product handling areas before restarting adjacent operations.

3.7 Blast Freezer

The blast freezer rapidly removes heat from the product to reach the required freezing condition. Stable compressor performance and air circulation are essential.

3.7.1 Compressor Failure

Affected parameters: temperature, air_speed_ms

Issue description: Compressor failure reduces cooling capacity and may also affect airflow performance.

Required operator action:

1. Clear the anomaly.
2. Stop the machine.
3. Start the machine.

Warning: If freezing performance does not recover promptly, isolate the unit and escalate.

3.8 Storage Freezer

The storage freezer preserves finished or semi-finished product under controlled low-temperature conditions. Door integrity directly affects temperature stability.

3.8.1 Door Stuck

Affected parameters: door_open, temperature

Issue description: Door stuck indicates that the access door is not opening or closing correctly, allowing unwanted temperature increase.

Required operator action:

1. Clear the anomaly.
2. Stop the machine.
3. Start the machine.

Note: Minimize door openings until the condition is fully resolved.

3.9 Cold Room

The cold room provides controlled refrigerated storage for ingredients or product in process. Temperature control and load distribution are necessary to maintain cooling efficiency.

3.9.1 Defrost Failure

Affected parameters: temperature

Issue description: Defrost failure causes ice accumulation and reduces heat exchange efficiency.

Required operator action:

1. Clear the anomaly.
2. Stop the machine.

3. Start the machine.

Caution: Repeated failures indicate the unit should be inspected before continued loading.

3.9.2 Overload

Affected parameters: occupancy_pct

Issue description: Overload means the room is holding more product than intended, restricting airflow and reducing cooling performance.

Required operator action:

1. Clear the anomaly.
2. Stop the machine.
3. Start the machine back to operation.

Note: Product arrangement should allow adequate air circulation throughout the room.

3.10 Batch Freezer

The batch freezer churns and freezes the product to the required texture and temperature. Motor performance and thermal control are both essential to complete the cycle.

3.10.1 Motor Failure

Affected parameters: dasher_rpm

Issue description: Motor failure prevents proper dasher rotation and interrupts the freezing and churning process.

Required operator action:

1. Clear the anomaly.
2. Stop the machine.
3. Start the machine.

Warning: If the motor fails again after restart, remove the unit from production.

3.10.2 Freeze Lock

Affected parameters: temperature

Issue description: Freeze lock indicates that the unit is excessively cold or mechanically restricted by frozen product, preventing normal continuation of the cycle.

Required operator action:

1. Clear the anomaly.
2. Stop the machine.
3. Start the machine.

Caution: Do not force continued operation if the chamber remains locked or abnormal resistance is observed.